Testing and Deploying Your API



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Coming Up



Options for testing API endpoints

- The HTTP_REPL
- .http-files in Visual Studio

Coming Up



Hosting and deploying an API

- Hosting on Azure
- Forwarded headers middleware
- Configuring logging on Azure with Application Insights
- Storing secrets in Azure Key Vault

Testing API Endpoints



Test for invalid input



Test for correct responses



Test your security

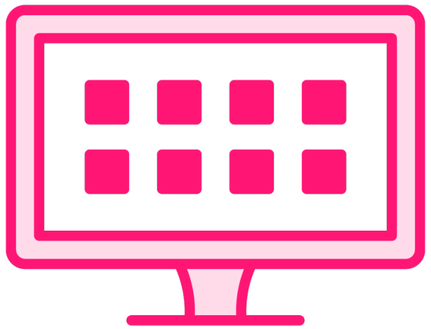


... and more

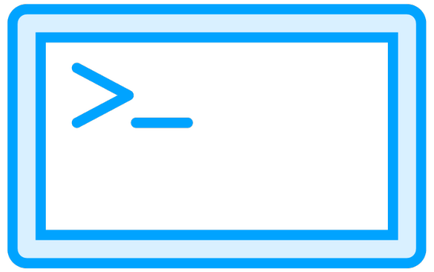
Testing API Endpoints



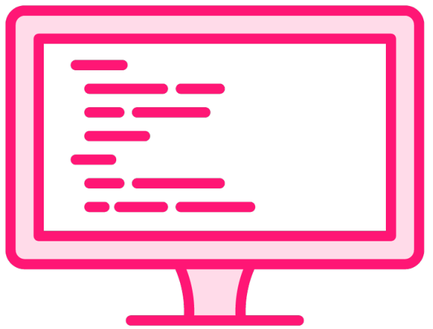
Swagger



Postman
(or comparable
alternatives)



The HTTP_REPL



.http files

Demo



Instaling the HTTP_REPL

Demo



Testing an API with the HTTP_REPL

Demo



Supporting authenticated requests when testing with the HTTP_REPL

Demo



Testing with .http files

Demo



Supporting authenticated requests when testing with .http files

Demo



Facilitating API testing with Endpoints Explorer

Publishing

Creating the set of files required to run the application



Deploying

Effectively copying the published files to the host



Hosting and Deploying an ASP.NET Core API

Local deployment:

- ProjectName/bin/{TypeOfBuild}/net8.0/

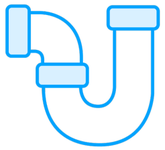
Local process manager:

- Kestrel (default)

Considering Your Deployment and Hosting Options



Scalability



Ease of maintenance



Cost



Application-specific requirements



...

Components like proxies and load balancers often obscure information about the request

Dealing with Proxies and Load Balancers

Information that's often obscured:

- Original scheme (http(s))
- Originating client IP

This information is forwarded in `X-Forwarded` headers

Demo



Configuring the forwarded headers middleware

Demo



Deploying to an Azure App Service

Demo



Configuring logging on Application Insights

Demo



Improving storing secrets with Azure Key Vault

Learning More About Deployments



Consider a more full-featured database server



Consider API management



Consider automating your Azure infrastructure



Consider automating your deployments

Summary



The HTTP_REPL (Read Eval Print Loop)

- A command-line tool for testing web APIs

.http files

- Integrated API testing approach straight from Visual Studio

Summary



Publishing

- Creating the set of files required to run the application

Deploying

- Copying the published files to the host

Summary



Deploying to an Azure App Service

- Forwarded headers middleware
- Configure logging on Application Insights
- Store secrets in Azure Key Vault

The End is Nigh...





Questions?
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You're ready
to be
AWESOME!